4.4.5 A model of computation

4.4.5.1 Turing machine

Content	Additional information
Be familiar with the structure and use of Turing machines that perform simple computations.	

<table>
<tr><th>Content</th><th>Additional information</th></tr>
<tr><td>Know that a Turing machine can be viewed as a computer with a single fixed program, expressed using:
a finite set of states in a state transition diagram
a finite alphabet of symbols
an infinite tape with marked-off squares
a sensing read-write head that can travel along the tape, one square at a time.
One of the states is called a start state and states that have no outgoing transitions are called halting states.</td><td>Exam questions will only be asked about Turing machines that have one tape that is infinite in one direction.</td></tr>
<tr><td>Understand the equivalence between a transition function and a state transition diagram.</td><td></td></tr>
<tr><td>Be able to:
represent transition rules using a transition function
represent transition rules using a state transition diagram
hand-trace simple Turing machines.</td><td></td></tr>
<tr><td>Be able to explain the importance of Turing machines and the Universal Turing machine to the subject of computation.</td><td>Turing machines provide a (general/formal) model of computation and provide a definition of what is computable.</td></tr>
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